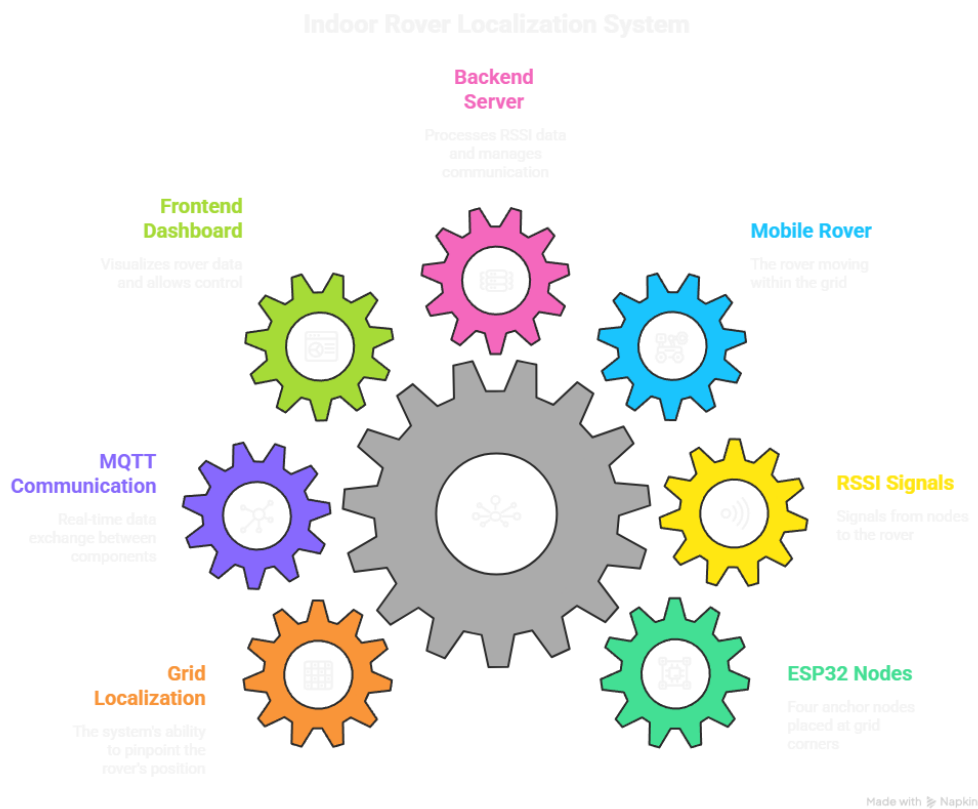
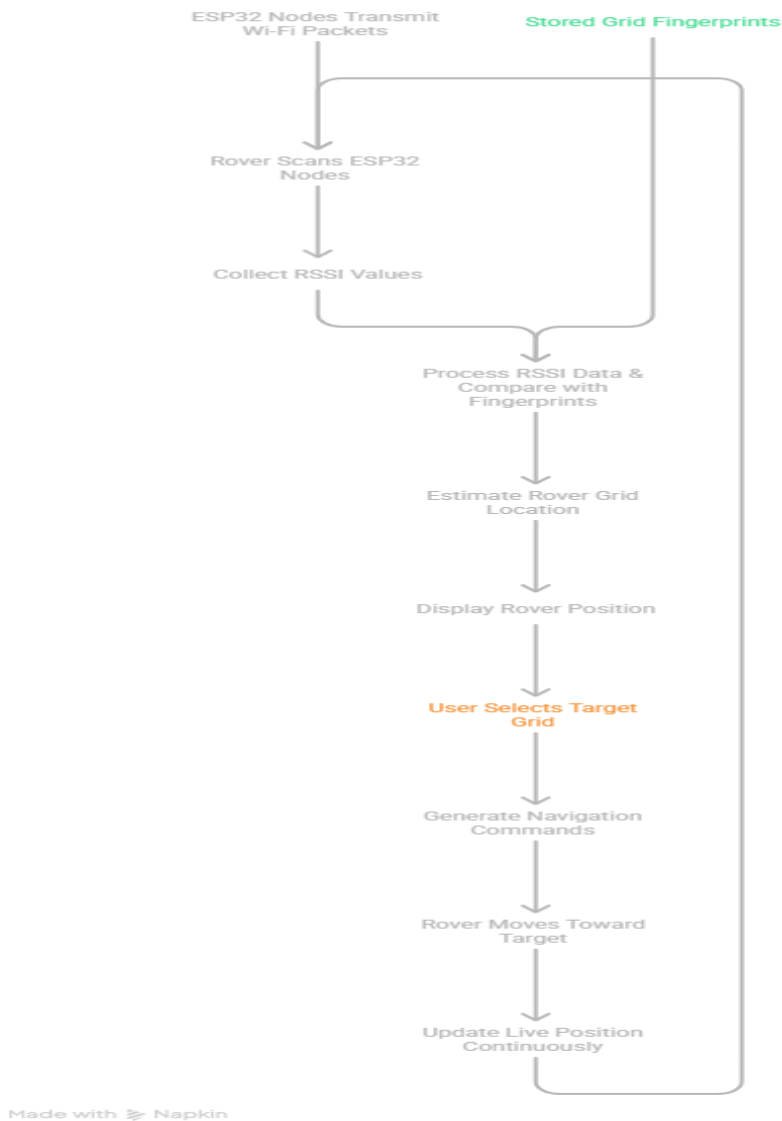




System Architecture

This diagram represents the overall architecture of our RSSI-based indoor localization system. Four ESP32 anchor nodes are placed at the corners of a selected indoor area and continuously transmit Wi-Fi packets. The rover collects RSSI values from these nodes and sends the data to the backend server through MQTT communication. The backend processes localization data and communicates with the frontend dashboard, allowing users to monitor rover movement, log RSSI fingerprints, and select target grids for autonomous navigation.

Indoor Rover Localization and Navigation Workflow



Localization and Navigation Workflow

This workflow shows how the indoor localization system operates in real time. ESP32 anchor nodes continuously broadcast Wi-Fi packets, and the rover scans these signals to collect RSSI values. The backend compares the collected RSSI patterns with stored grid fingerprints to estimate the rover's approximate position. Through the frontend dashboard, users can monitor rover movement and select destination grids, after which the backend generates navigation commands for autonomous rover movement.

